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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
18/163,169	02/01/2023	Hayley WARSINSKE	49407-752.301	5871

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EXAMINER
NEGIN, RUSSELL SCOTT

ART UNIT	PAPER NUMBER
1672	

NOTIFICATION DATE	DELIVERY MODE
06/27/2024	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

patentdocket@wsgr.com

Office Action Summary	Application No. 18/163,169	Applicant(s) WARSINSKE et al.	
	Examiner RUSSELL S NEGIN	Art Unit 1672	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 04 June 2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☐ This action is **FINAL.** 2b) ☑ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1,3-10,12,14,16-18 and 20-24 is/are pending in the application.

5a) Of the above claim(s) 3-6 and 8-10 is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1,7,12,14,16-17 and 20-24 is/are rejected.

8) ☑ Claim(s) 18 is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☐ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

DETAILED ACTION

Comments

The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA. In the event the determination of the status of the application as subject to AIA 35 U.S.C. 102 and 103 (or as subject to pre-AIA 35 U.S.C. 102 and 103) is incorrect, any correction of the statutory basis (i.e., changing from AIA to pre-AIA) for the rejection will not be considered a new ground of rejection if the prior art relied upon, and the rationale supporting the rejection, would be the same under either status.

Claims 3-6 and 8-10 are withdrawn from further consideration pursuant to 37 CFR 1.142(b) as being drawn to a nonelected species, there being no allowable generic or linking claim. Election was made **without** traverse in the reply filed on 15 May 2023.

Claims 1, 3-10, 12, 14, 16-18, and 20-24 are pending in the application.

Claims 1, 7, 12, 14, 16-18, and 20-24 are examined in the instant Office action.

Withdrawn Rejections

The rejections under 35 U.S.C. 112 and 35 U.S.C. 103 are withdrawn in view of amendments filed to the instant set of claims on 4 June 2024.

Claim Rejections - 35 USC § 101

35 U.S.C. 101 reads as follows:

> Whoever invents or discovers any new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof, may obtain a patent therefor, subject to the conditions and requirements of this title.

The following rejection is newly applied:

Claim(s) 1, 7, 12, 14, 16-17, and 20-24 is/are rejected under 35 U.S.C. 101 because the claimed invention is directed to an abstract idea/law of nature/natural phenomenon without significantly more. The claims are drawn to methods.

In accordance with MPEP § 2106, claims found to recite statutory subject matter (***Step 1*** **: YES**) are then analyzed to determine if the claims recite any concepts that equate to an abstract idea, law of nature or natural phenomenon (*Step 2A, Prong 1*). In the instant application, the claims recite the following limitations that equate to an abstract idea:

- The independent claims recite detecting colorectal cancer by measuring the amounts of at least three proteins from a given list.
- The independent claims recite using machine learning that differentiates between control subjects, subjects with benign polyps, and subjects with advanced adenoma to detect an increased risk of colorectal cancer.
- The independent claims recite detecting colorectal cancer in patients with increased risk by using a colonoscopy.
- Claims 7 and 23 constrain the protein panel.
- Claim 12 constrains the type of colorectal cancer.
- Claims 14 and 22 constrain the type of biological sample.
- Claim 16 constrains the type of machine learning model.
- Claim 17 stratifies the stages of colorectal cancer.
- Claim 24 constrains the training data.

These recitations are similar to the concepts of collecting information, analyzing it and displaying certain results of the collection and analysis in *Electric Power Group, LLC, v. Alstom* (830 F.3d 1350, 119 USPQ2d 1739 (Fed. Cir. 2016)), organizing and manipulating information through mathematical correlations in *Digitech Image Techs., LLC v Electronics for Imaging, Inc.* (758 F.3d 1344, 111 U.S.P.Q.2d 1717 (Fed. Cir. 2014)) and comparing information regarding a sample or test to a control or target data in *Univ. of Utah Research Found. v. Ambry Genetics Corp.* (774 F.3d 755, 113 U.S.P.Q.2d 1241 (Fed. Cir. 2014)) and *Association for Molecular Pathology v. USPTO* (689 F.3d 1303, 103 U.S.P.Q.2d 1681 (Fed. Cir. 2012)) that the courts have identified as concepts that can be practically performed in the human mind or mathematical relationships. Therefore, these limitations fall under the "Mental process" and "Mathematical concepts" groupings of abstract ideas. Merely reciting that a mental process is being performed in a generic computer environment does not preclude the steps from being performed practically in the human mind or with pen and paper as claimed. If a claim limitation, under its broadest reasonable interpretation, covers performance of the limitation in the mind but for the recitation of generic computer components, then if falls within the "Mental processes" grouping of abstract ideas. As such, claim(s) 1, 7, 12, 14, 16-17, and 20-24 recite(s) an abstract idea/law of nature/natural phenomenon (***Step 2A, Prong 1*** **: YES**).

Claims found to recite a judicial exception under *Step 2A, Prong 1* are then further analyzed to determine if the claims as a whole integrate the recited judicial exception into a practical application or not (*Step 2A, Prong 2*). This judicial exception is not integrated into a practical application because the claims do not recite an additional

element that reflects an improvement to technology or applies or uses the recited judicial exception to effect a particular treatment for a condition. Rather, the instant claims recite additional elements that amount to mere instructions to implement the abstract idea in a generic computing environment or mere instructions to apply the recited judicial exception via a generic treatment. While the claims recite performing colonoscopies on patients with increased risk of colorectal cancer, these colonoscopies are performed for the purpose of detecting (and not treating) colorectal cancer.

As such, these limitations equate to mere instructions to implement the abstract idea on a generic computer that the courts have stated does not render an abstract idea eligible in *Alice Corp.,* 573 U.S. at 223, 110 USPQ2d at 1983. See also 573 U.S. at 224, 110 USPQ2d at 1984. As such, claims 1, 7, 12, 14, 16-17, and 20-24 is/are directed to an abstract idea/law of nature/natural phenomenon (***Step 2A, Prong 2* : NO**).

Claims found to be directed to a judicial exception are then further evaluated to determine if the claims recite an inventive concept that provides significantly more than the judicial exception itself (*Step 2B*). The claims do not include additional elements that are sufficient to amount to significantly more than the judicial exception because the claims recite additional elements that equate to mere instructions to apply the recited exception in a generic way or in a generic computing environment. While the claims recite assaying samples, the prior art of record (e.g. Vogelstein et al., Janjua et al.) demonstrates that assaying samples for the purpose of identifying a risk of colorectal cancer is routine and conventional in the prior art.

Claims that amount to nothing more than an instruction to apply the abstract idea using a generic computer do not render an abstract idea eligible. *Alice Corp.,* 573 U.S.

at 223, 110 USPQ2d at 1983. See also 573 U.S. at 224, 110 USPQ2d at 1984. MPEP 2106.05(f) discloses that mere instructions to apply the judicial exception cannot provide an inventive concept to the claims. The additional elements do not comprise an inventive concept when considered individually or as an ordered combination that transforms the claimed judicial exception into a patent-eligible application of the judicial exception. Therefore, the claims do not amount to significantly more than the judicial exception itself (***Step 2B*** *: No*). As such, claims 1, 7, 12, 14, 16-17, and 20-24 is/are not patent eligible.

Allowable Subject Matter

Claim 18 is objected to as being dependent upon a rejected base claim, but would be allowable if rewritten in independent form including all of the limitations of the base claim and any intervening claims.

Claim 18 is subject matter eligible because the claim recites the practical application of a particular treatment of colorectal cancer.

The claims are free of the prior art because the prior art does not teach performing machine learning to differentiate between control subjects, subjects with advanced adenomas, and subjects with benign tumors.

Related Prior Art

The document of Singh et al. [Cancer Cell, volume 1, 2002, pages 203-209; on IDS] teaches using specific genes as biomarkers and machine learning to detect

prostate cancer. However, Singh et al. does not teach the recited protein panel and pertains to prostate cancer (and not colorectal cancer).

The document of Golub et al. [Science, volume 286, 1999, pages 531-537; on IDS] teaches molecular classification of cancer using class discovery and class prediction by gene expression monitoring. However, Golub et al. pertains to leukemia (and not colorectal cancer) and does not teach the recited protein panel.

E-mail Communications Authorization

Per updated USPTO Internet usage policies, Applicant and/or applicant's representative is encouraged to authorize the USPTO examiner to discuss any subject matter concerning the above application via Internet e-mail communications. See MPEP 502.03. To approve such communications, Applicant must provide written authorization for e-mail communication by submitting the following statement via EFS-Web (using PTO/SB/439) or Central Fax (571-273-8300):

> Recognizing that Internet communications are not secure, I hereby authorize the USPTO to communicate with the undersigned and practitioners in accordance with 37 CFR 1.33 and 37 CFR 1.34 concerning any subject matter of this application by video conferencing, instant messaging, or electronic mail. I understand that a copy of these communications will be made of record in the application file.

Written authorizations submitted to the Examiner via e-mail are **NOT** proper. Written authorizations must be submitted **via EFS-Web** (using PTO/SB/439) **or Central Fax** (571-273-8300). A paper copy of e-mail correspondence will be placed in the patent application when appropriate. E-mails from the USPTO are for the sole use of the intended recipient, and may contain information subject to the confidentiality requirement set forth in 35 USC § 122. See also MPEP 502.03.

Conclusion

No claim is allowed.

Any inquiry concerning this communication or earlier communications from the Examiner should be directed to Russell Negin, whose telephone number is (571) 272-1083. This Examiner can normally be reached from Monday through Thursday from 8 am to 3 pm and variable hours on Fridays.

If attempts to reach the Examiner by telephone are unsuccessful, the Examiner's Supervisor, Larry Riggs, Supervisory Patent Examiner, can be reached at (571) 270-3062.

/RUSSELL S NEGIN/
Primary Examiner, Art Unit 1672
23 June 2024